THEATERFORGE LLC

CAPABILITY STATEMENT
Rev. 2026-10-01

Deterministic Multi-Domain Simulation and Trade Study Software

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COMPANY DATA

LEGAL NAME
ThEaterForge LLC

ENTITY TYPE
Limited Liability Company

STATE OF FORMATION
Utah

STATE ENTITY NO.
14733139-0160

ESTABLISHED
2 September 2026

BUSINESS SIZE
Small business

TECH READINESS
TRL 4

UEI
FTQ7GVNLKDX5

CAGE CODE
257P5

PLACE OF PERFORMANCE
United States

NAICS CODES

541715	R&D in the Physical, Engineering, and Life Sciences (primary)
541511	Custom Computer Programming Services
541512	Computer Systems Design Services
541330	Engineering Services

PSC CODES

AC12	R&D Defense System: Applied Research
DA01	IT and Telecom: Business Application Software Development

POINT OF CONTACT

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CORE COMPETENCIES

- **Deterministic simulation engineering.** Multi-domain theater models - space, air, maritime, ground - executing as reproducible experimental instruments rather than visualizations. Identical inputs produce byte-identical outputs, verified by hashing.
- **Automated trade space exploration.** Design-of-experiments harnesses screening large parameter spaces to identify driving factors. Full factorial, saturated two-level screening, and Latin hypercube designs, with main-effects analysis and pass/fail verdicts against stated requirements.
- **Browser-native scientific computing.** High-performance physics and analysis executing entirely client-side: no install, no server-side compute, no runtime license, no administrative rights required on the host workstation.
- **Analyst-facing decision support tooling.** 3D geospatial and 2D tactical visualization, scenario authoring, and self-contained exportable reporting.

DIFFERENTIATORS

- **Zero-footprint deployment.** Runs in a standard browser on the machine the analyst already has. No installation, no backend to accredit, no workstation configuration - materially lowering the barrier to fielding analysis capability inside restricted enclaves.
- **Air-gapped by default.** Ships with its network gate closed. Simulation data never leaves the host machine under any configuration: no telemetry, no scenario upload, no external analysis service.
- **Findings a nominal run reports as success.** In a layered air and missile defense study, a raid of eight was fully intercepted with two tenths of a second of worst-case margin remaining - passing the leak-rate requirement and failing the timing requirement in the same run. In a multi-orbit missile-warning study, operator decision delay dominated every architecture parameter by roughly six to one. Both results are committed, hashed, and regenerable by a third party. Both are results about the model, not about any fielded architecture: parameter values are notional and the sensing models are not yet validated. What is demonstrated is the method and the reproducibility of the finding - whether a ranking of this kind holds as model fidelity changes is an open question, and is the question Phase I work is designed to answer.
- **Provable reproducibility.** Every probabilistic outcome is keyed to the identity of the individual event rather than a global sequence. Adding an element to a scenario does not perturb unrelated outcomes - the property that makes a measured difference between design points attributable to the design, not to noise.
- **One engine, two modes.** The interactive console and the batch analysis harness execute the same code on the same physics cadence. No data-generation versus presentation divergence to reconcile.
- **Screening scale, verified.** A 30-parameter trade space is screened in 32 runs, against 1.07 billion for the equivalent full factorial. The design is verified by automated test to be saturated, to keep every pair of columns orthogonal, to balance every column, and to recover a known effect from a synthetic response. Test output is committed and hashed.
- **Throughput.** Approximately 1,000-3,000x real time on a single laptop core, scenario-dependent. A twelve-design-point study covering six engagement-hours completes in roughly ten seconds.
- **Engineering discipline evidenced, not asserted.** 379 automated tests across 36 suites at tag v0.3.0 (2026-09-26), static type checking over both the application and the command-line analysis harness, generated software bill of materials, continuous integration, and a runnable timestep convergence study that reproduces its own verification result on demand.
- **Honest verification posture.** The system is verified, and one component - orbital propagation including J2 secular perturbation - is validated against published reference values. Remaining components are documented as unvalidated. Byte-identical reproduction is claimed for the same code on the same runtime, and every artifact records the version used. Capability and limitation are stated together.

PAST PERFORMANCE AND PRINCIPAL EXPERIENCE

ThEaterForge LLC has no federal prime or subcontract performance to date. The following is the relevant experience of the principal.

- **ThEaterForge simulation engine - sole architect and developer.** Multi-domain theater engine with integrated trade study harness. Delivered ballistic missile defense, integrated air and missile defense layering, and naval saturation attack scenario classes.
- **Data visualization consultant, LMI - three years.** Analytical and decision-support data visualization within a professional services organization serving federal government clients.

SECURITY AND DATA POSTURE

No classified data has been used in or is present in the system. Platform and weapon performance figures are compiled from unclassified open-source references combined with modeling judgment, and are structured as swappable configuration - controlled figures can be substituted without modifying engine logic.

ThEaterForge LLC is a small business seeking SBIR/STTR Phase I opportunities, subcontract roles supporting modeling and simulation efforts, and prototype development under OTA or similar instruments.